



2016 中国互联网安全大会
China Internet Security Conference

协同联动 共建安全+命运共同体

Security and Privacy in Mobile Advertising

Hao Chen

University of California,
Davis

Ads are the Financial Pillar of Mobile Computing



中国互联网安全大会



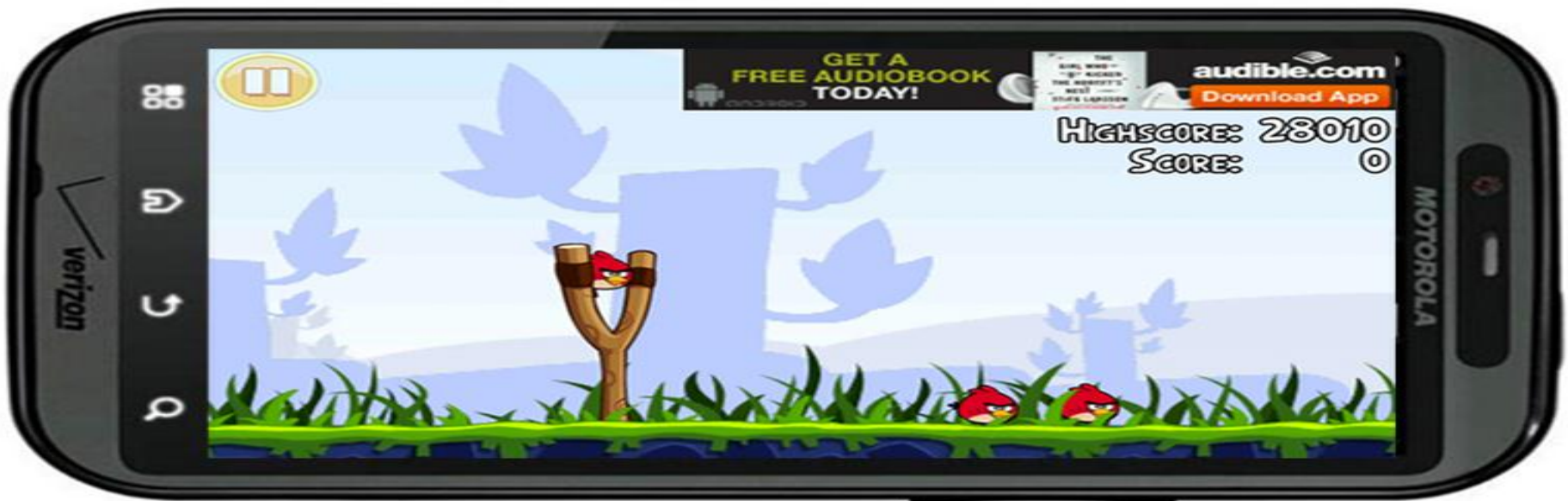
360互联网安全中心

Most mobile software is free.

- Android
- Most apps

Where does revenue come?

- In-app purchase
- Ads



Parties

- Users
- App developers
- Ad providers
- Advertisers

Security Problems



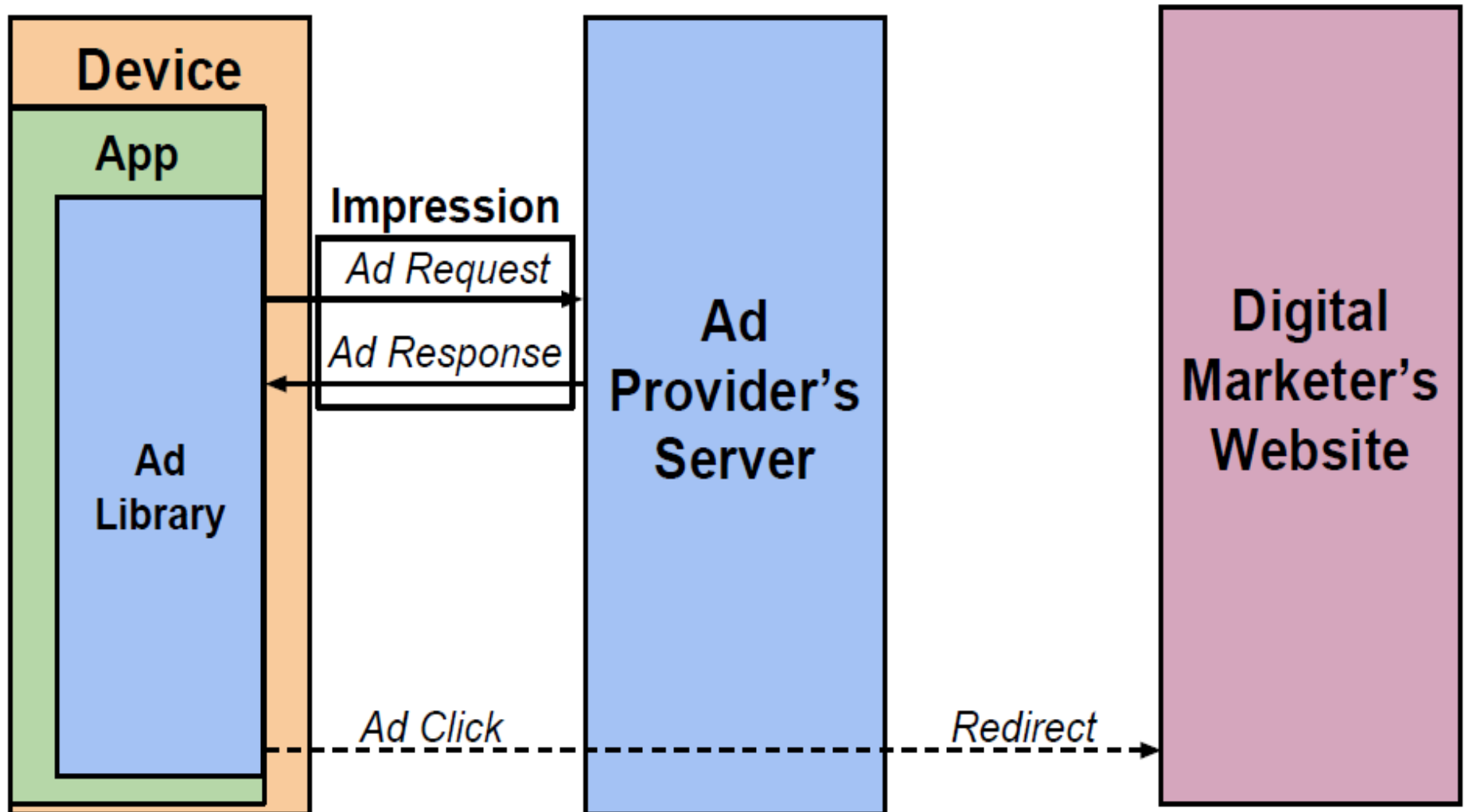
中国互联网络安全大会



360互联网安全中心

- App developers: ad revenue diversion due to cloning
- Ad providers / advertisers: ad fraud

Anatomy of Mobile Ad Traffic



Cloning harms app ecosystem

- Developers: lose revenue and incentives to make apps
- Markets: polluted search results
- Users: difficult to find high quality apps

Goals

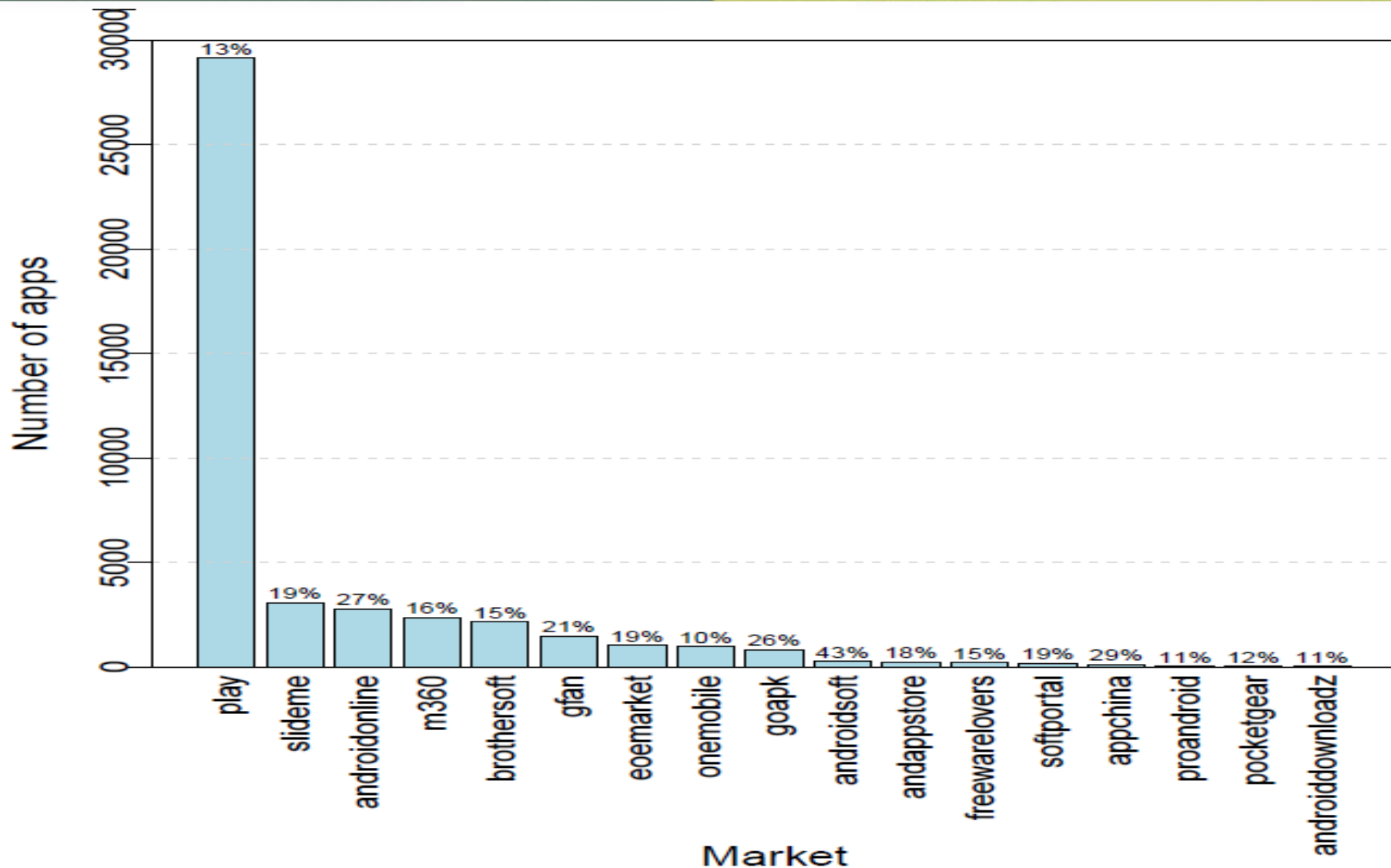
Characterize cloned apps

- Market
- App category
- Ad provider

Quantify impact on developers

- Loss of ad revenue
- Loss of user base

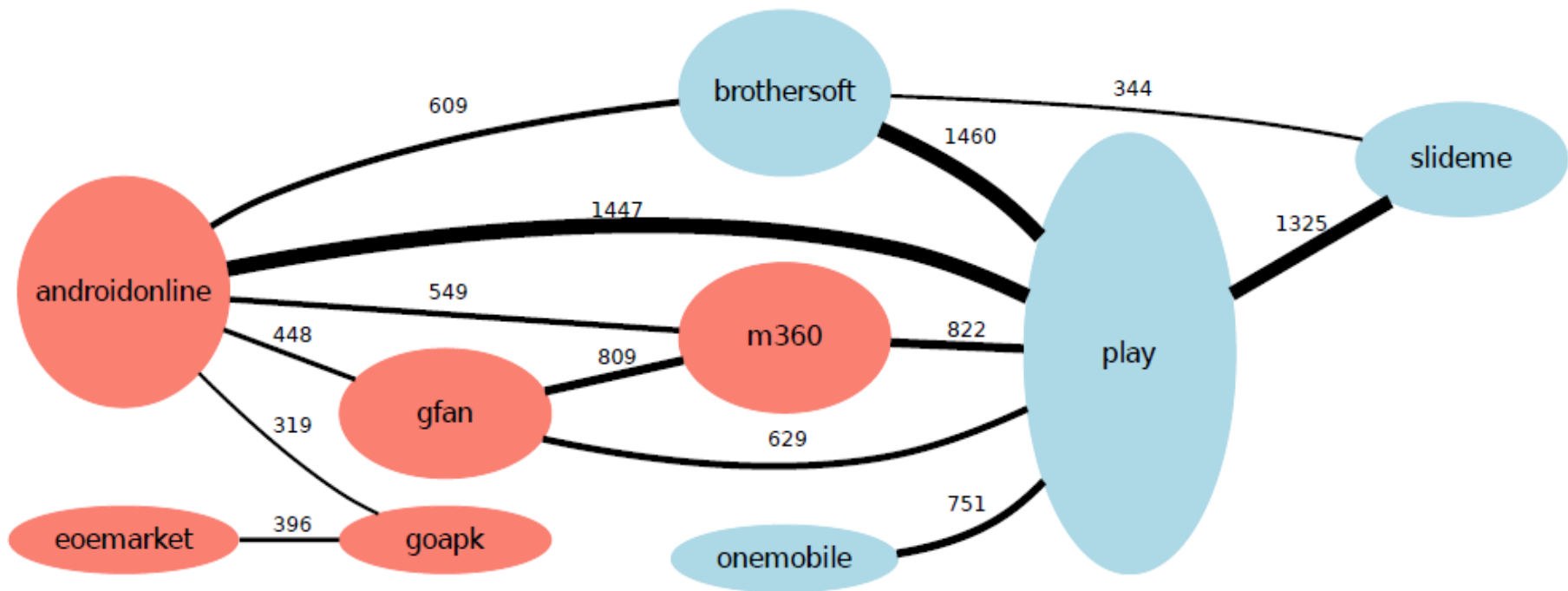
Dataset: 265,000 apps from 17 markets



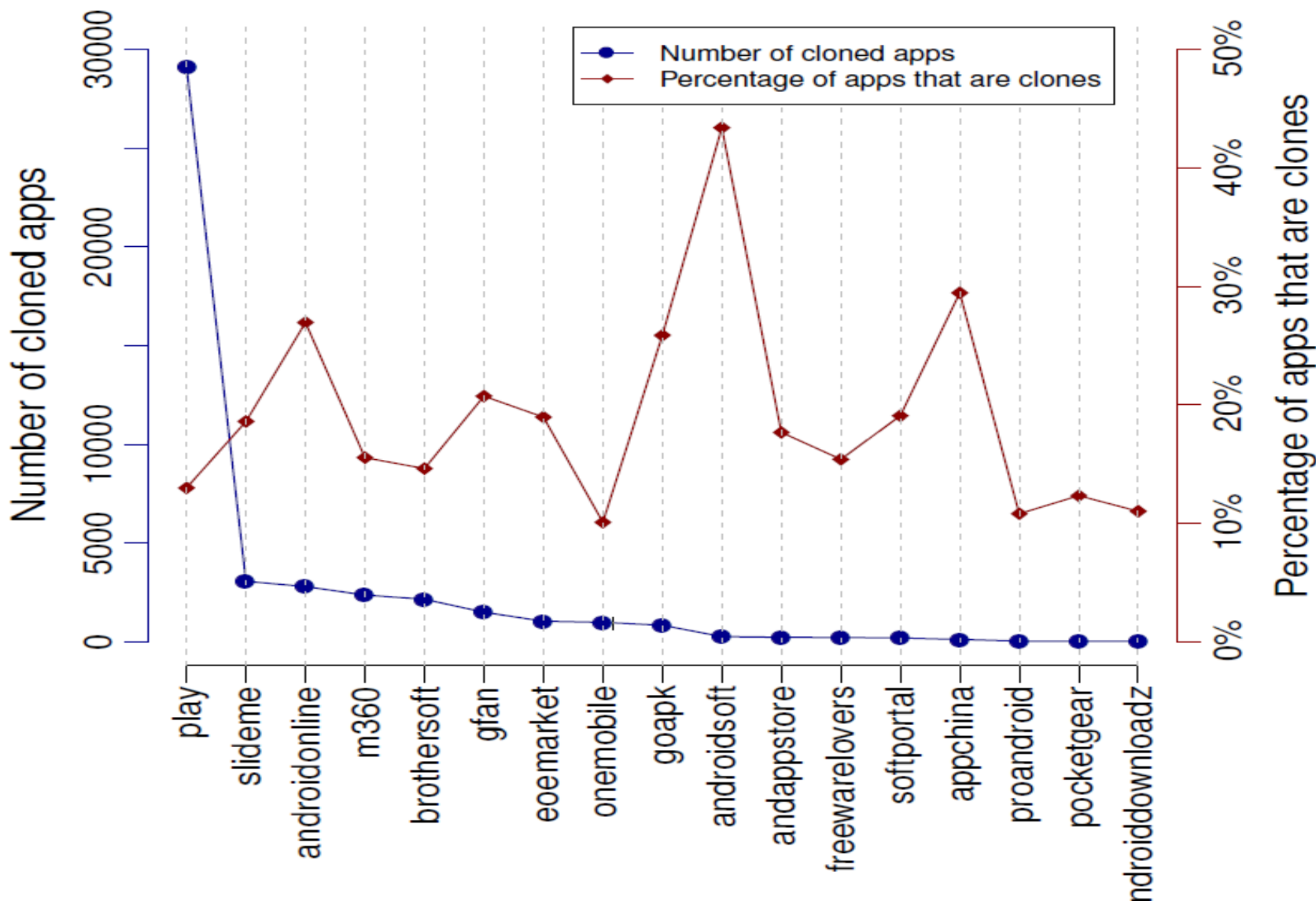
Cloning between Markets

Found

- >5,000 clusters of clones
- >44,000 unique apps



Cloning on Each Market



Determine Impact



中国互联网安全大会



360互联网安全中心

- Naive question
How many times has an app been cloned?
- Better question
How many people use a cloned app instead of the original?

How to Determine Impact



中国互联网安全大会



360互联网安全中心

- Count the users running each app.
- Determine the ownership of each app.
- Classify original vs. plagiarized apps.

Count Users Running Each App



中国互联网安全大会



360互联网安全中心

Potential vantage points

- On the device
- On the app server
- On the network

Count Users Running Each app



中国互联网安全大会



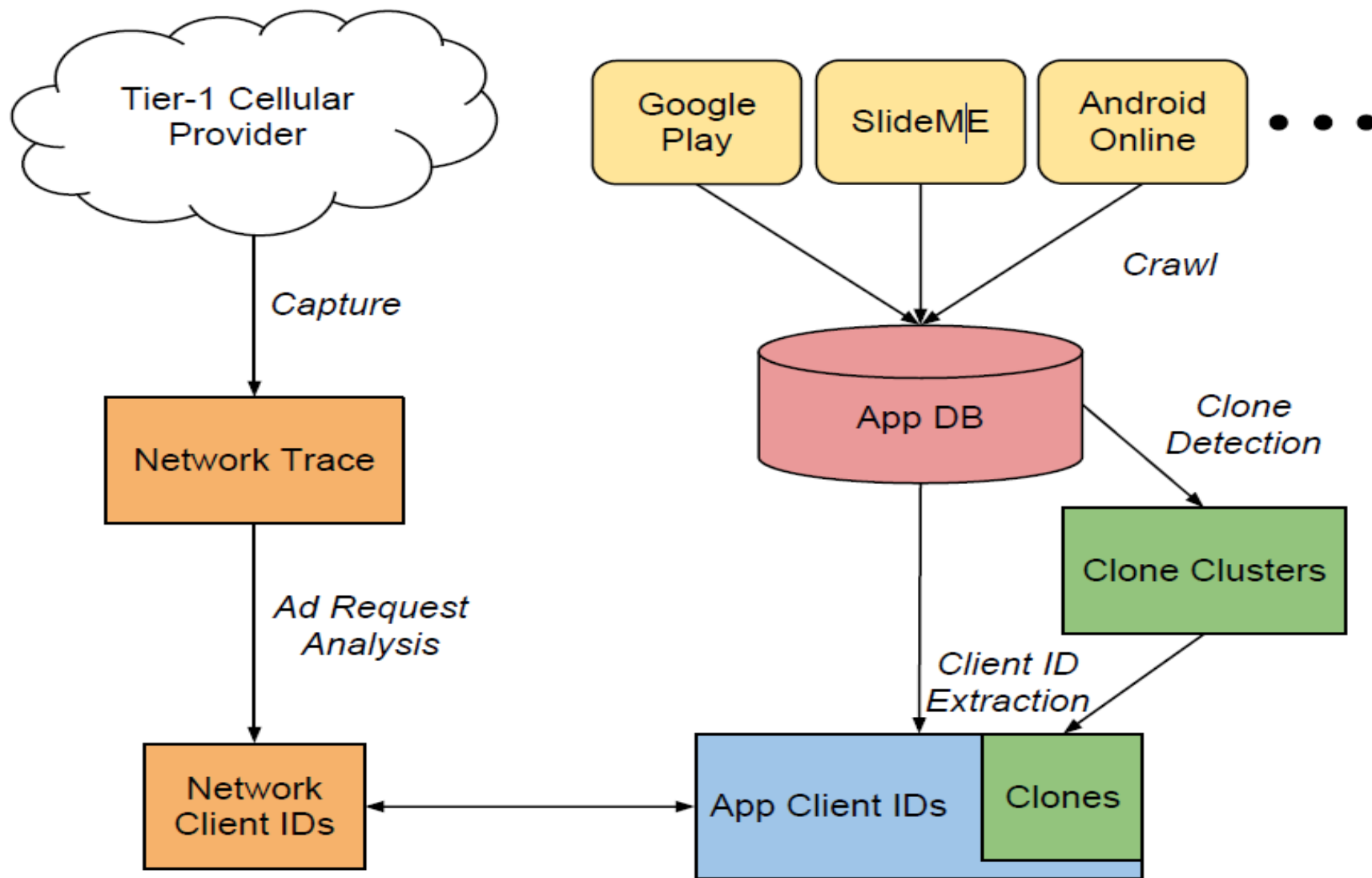
Combine traffic analysis with static analysis

On the network

- Capture ad traffic
- Extract client IDs

In the lab, for each app

- Extract client ID
- Determine whether it is original or clone



Traffic Analysis



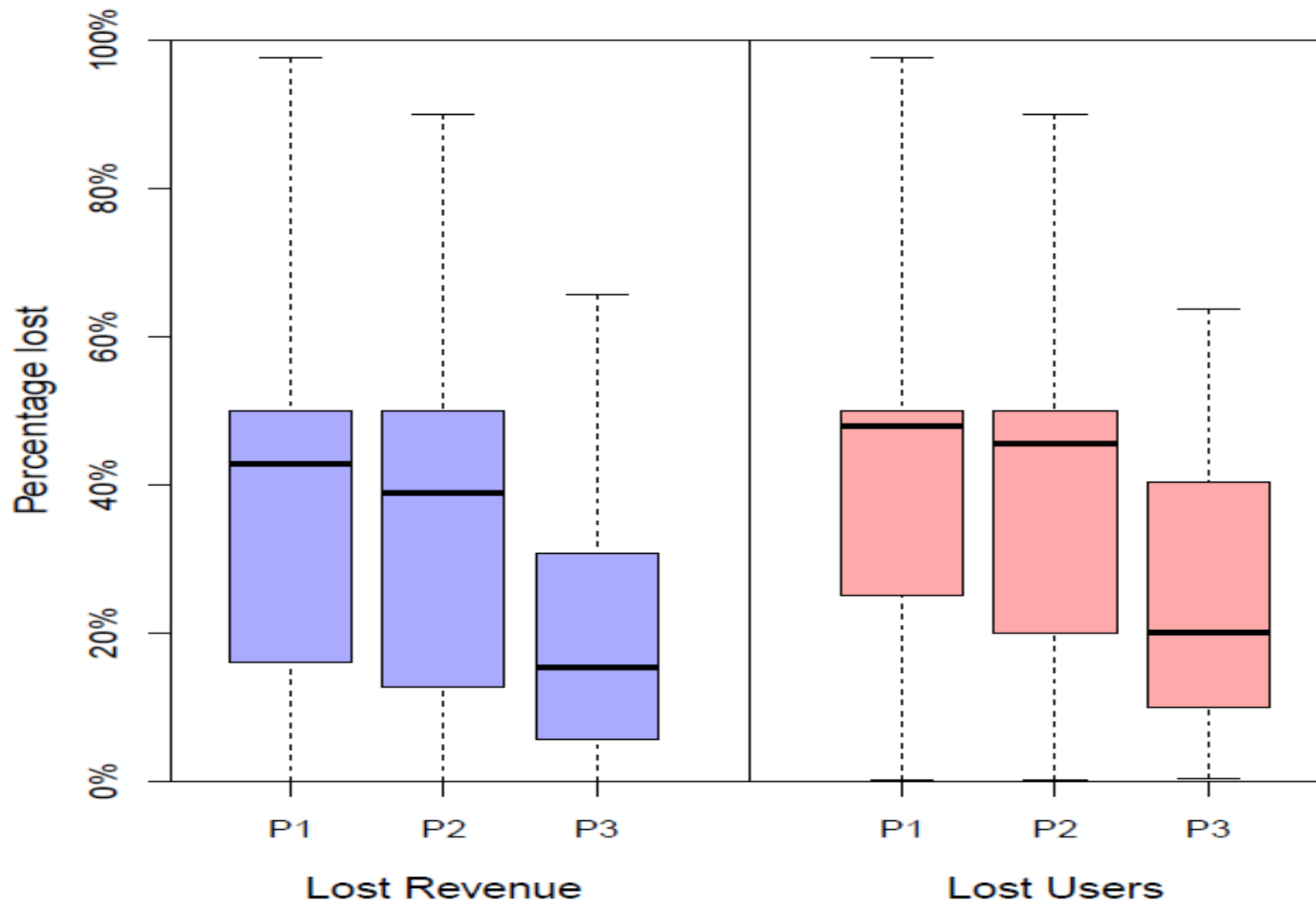
中国互联网安全大会



360互联网安全中心

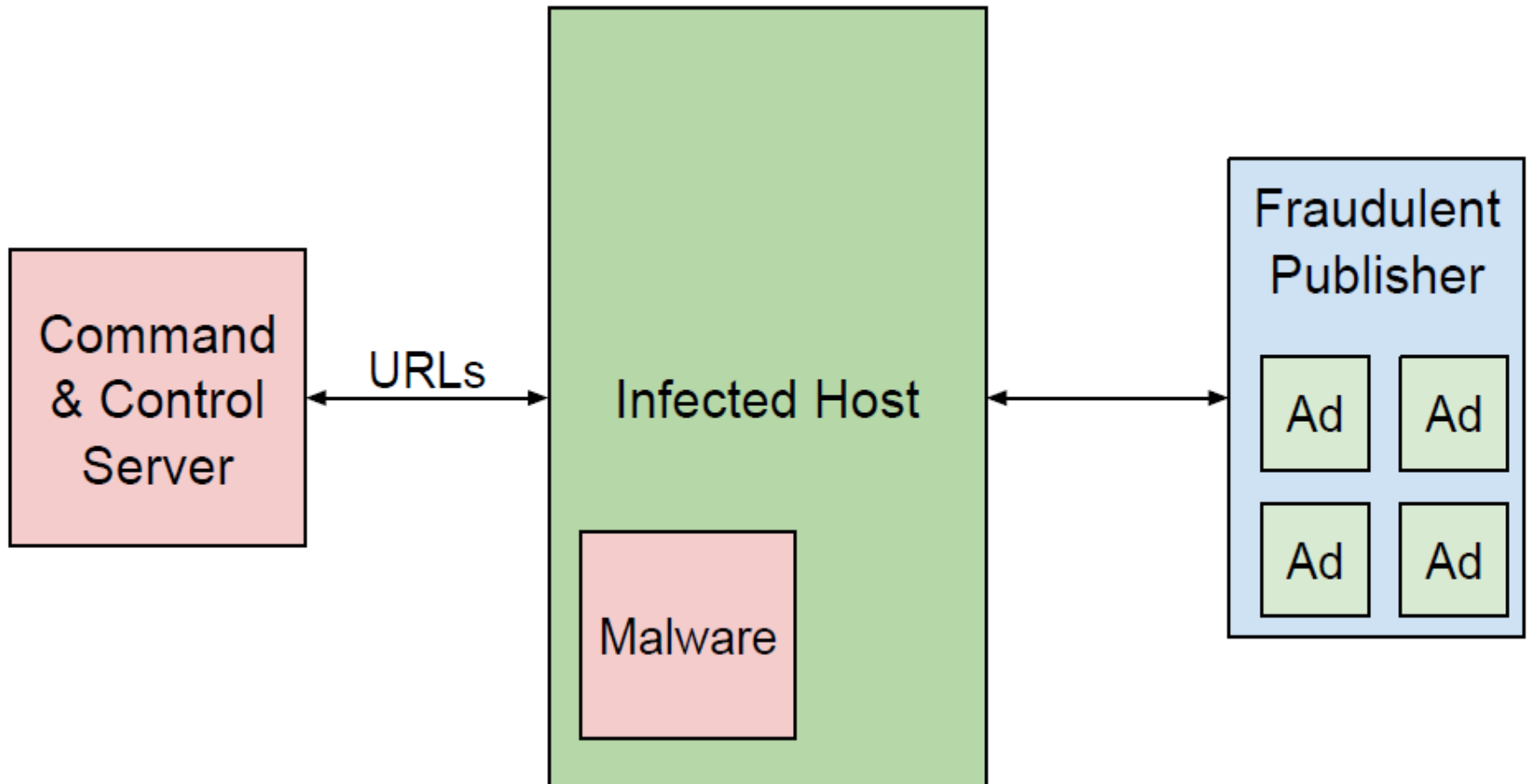
- Collaborated with a major US cellular provider.
- Captured 2.6 billion packets in 12 days.
- Removed all user-identifying information.

Loss of Revenue and User Base



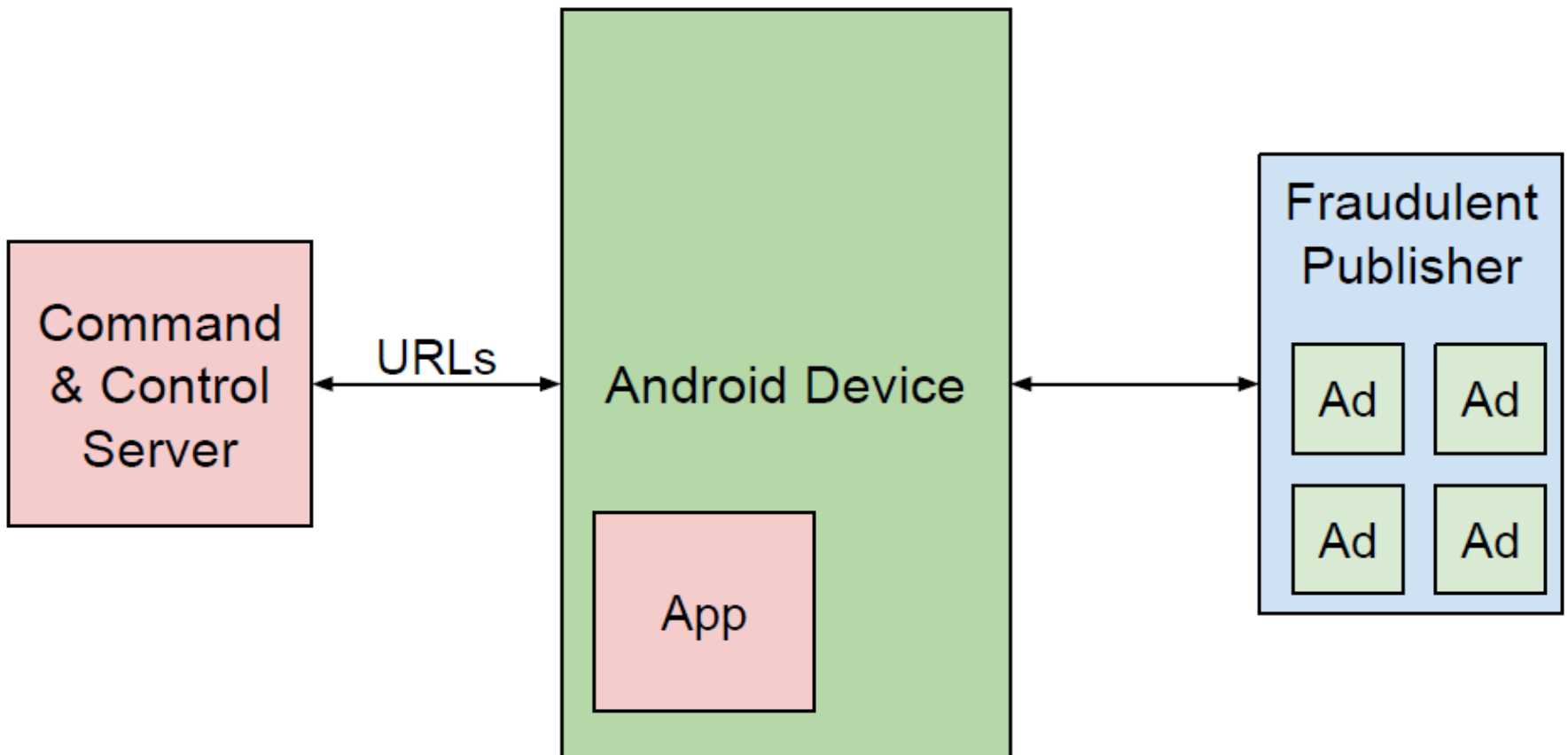
Web Ad Fraud

Programs that automatically “view” ads and “click” them



Mobile Ad Fraud

Apps that automatically “view” ads and “click” them



Goals

- Design system for automatically detecting ad traffic
- Use system to detect fraud and other undesirable ad behavior

Ad Fraud



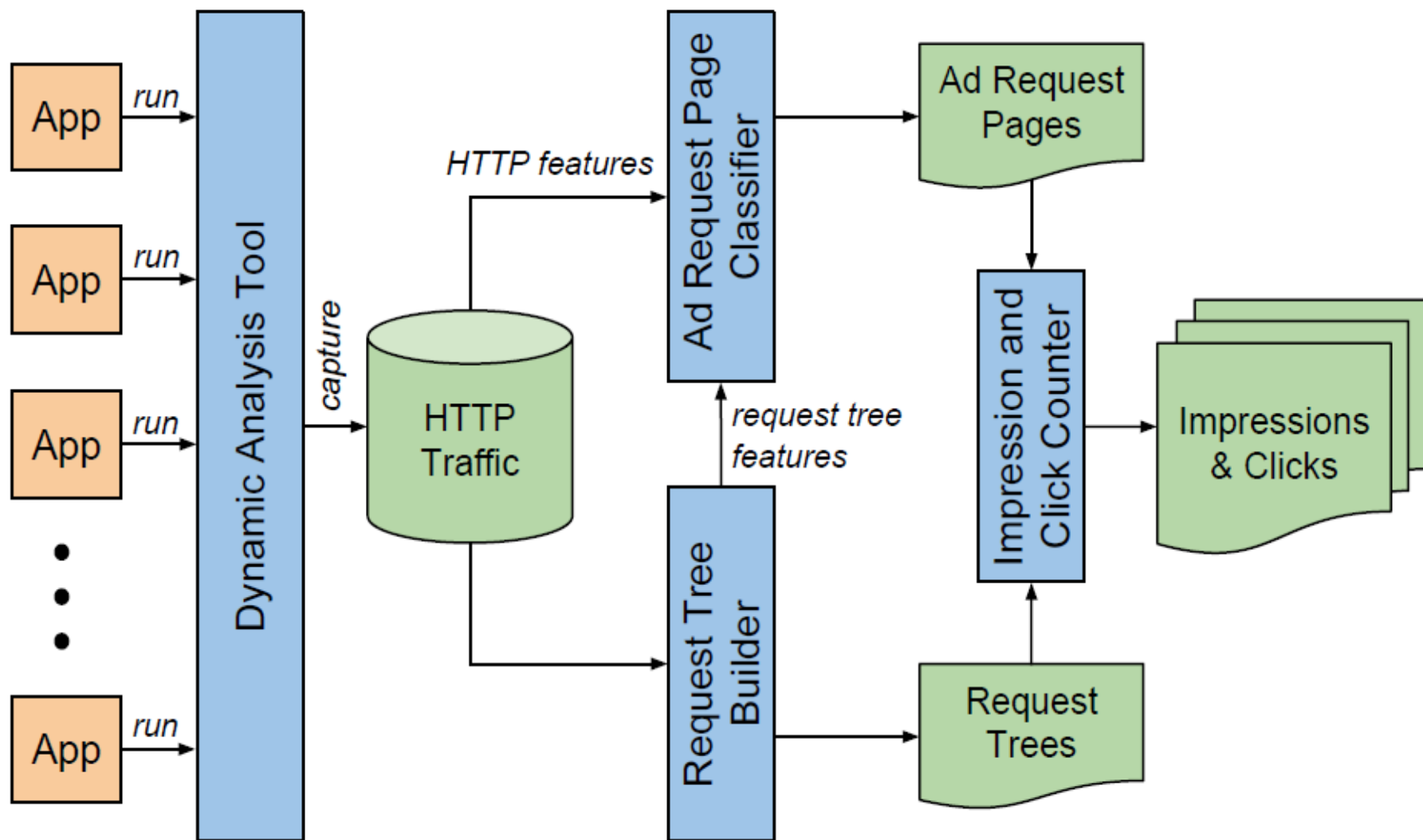
中国互联网络安全大会



360互联网安全中心

- Impression fraud: request ads when in the background.
- Click fraud: click on ads without user interaction.

MAdFraud



For each app:

- Create an emulator and install the app
- Run app in the foreground for 1 minute
- Use an Intent to move the app into the background
- Run app in the background for 1 minute
- Capture all network traffic throughout

Never interact with app to ensure all detected clicks are fraudulent

Identify Impressions and Clicks

Simple approach

- Reverse engineer ad libraries to find ad request and click URLs
- Use URLs to count the number of impressions and clicks

Problems

- Time intensive for the many ad libraries (80+)
- Lack context
 - Was the ad resold?
 - Was the click proceeded by an impression?

Automatically Identify Impressions and Clicks



中国互联网络安全大会



Build request trees

Identify impressions using machine learning based on

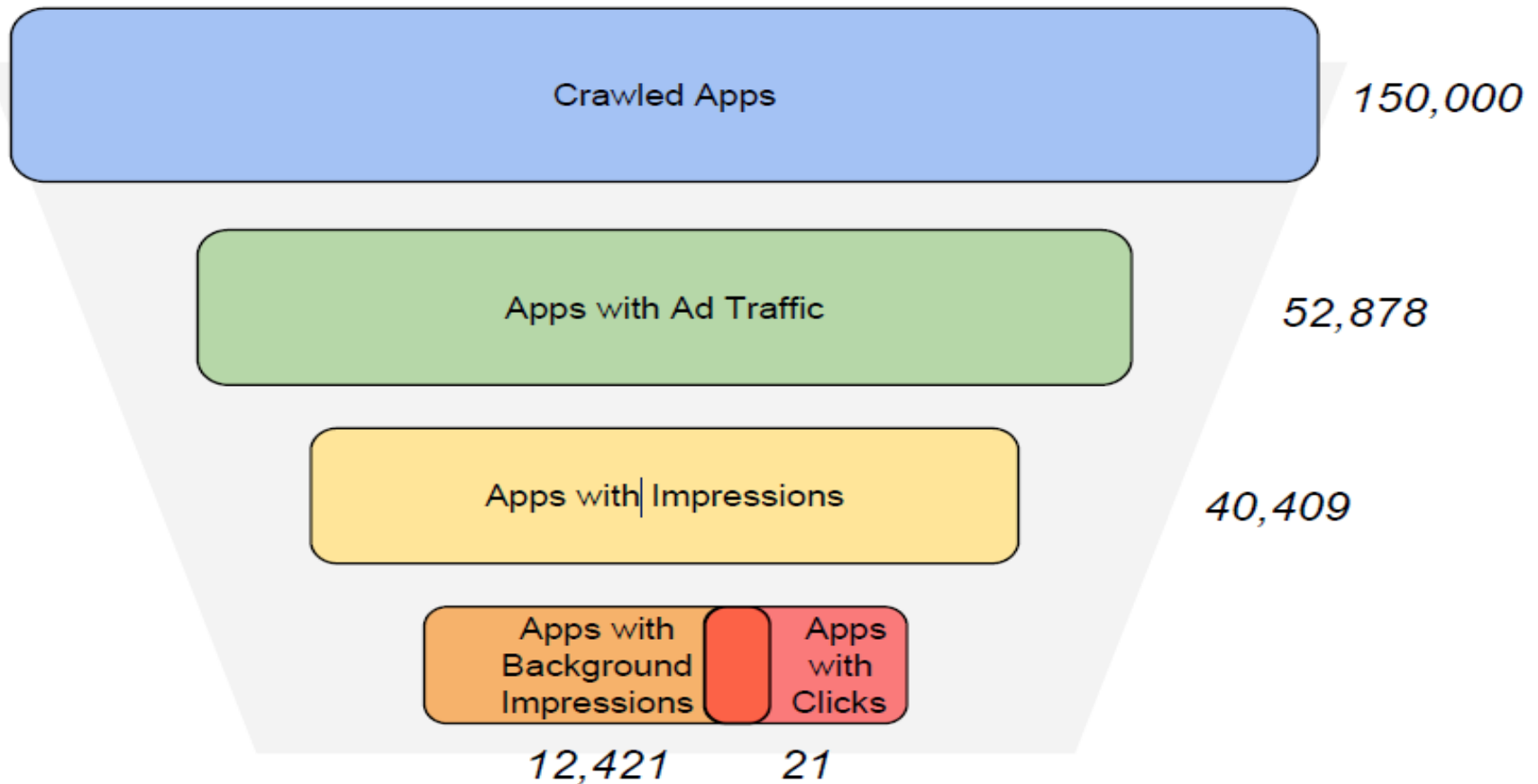
- Request tree properties
- Query parameters
- HTTP properties

Identify clicks based on request trees and impressions.

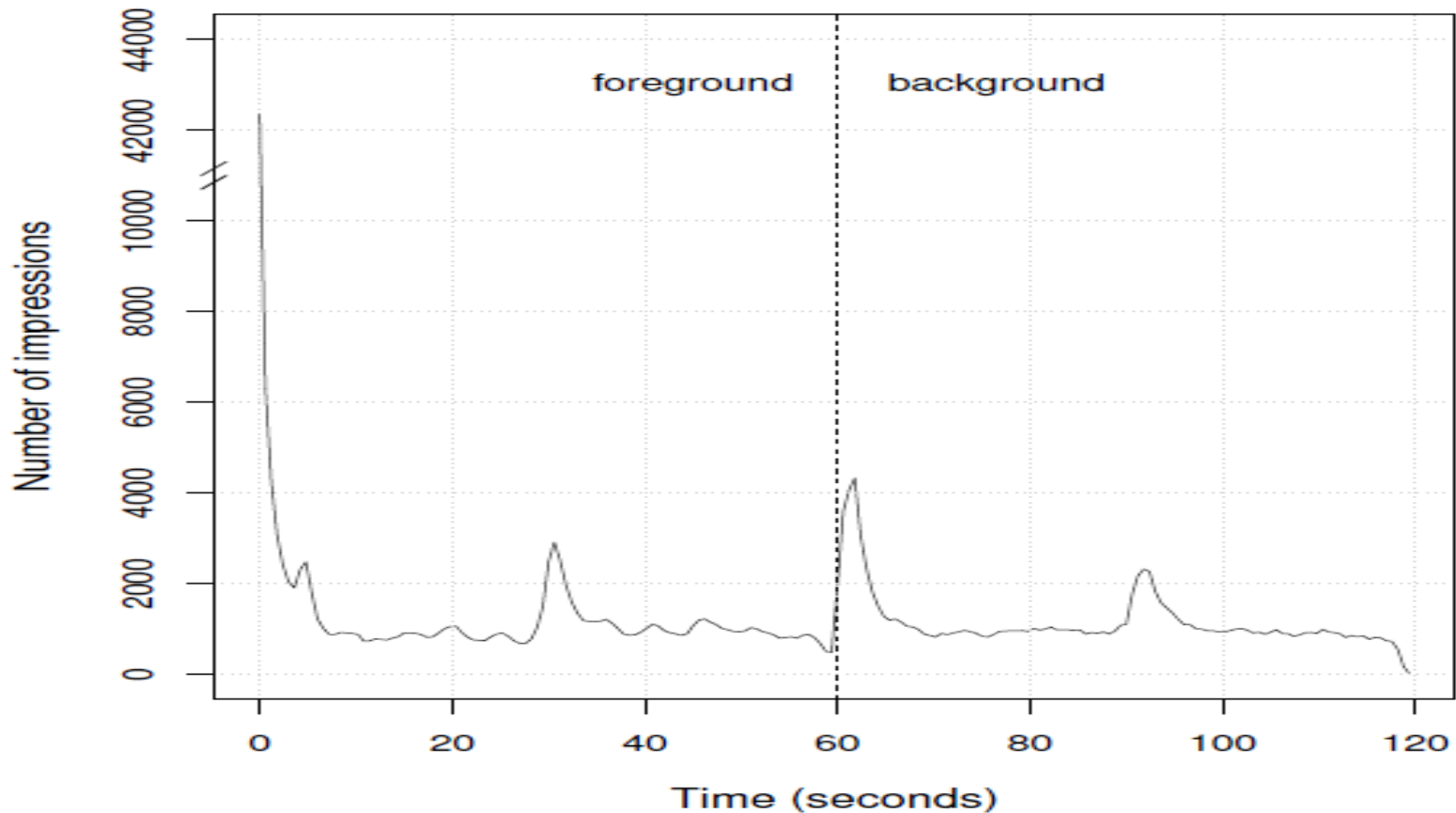
Dataset and Results Overview

Dataset

150,000 apps crawled from 19 markets



Finding: Background Impressions



12,421 apps make impressions in the background

Finding: Click Fraud

App	# Impressions	# Clicks	Click Interval (s)	Ad providers
79b85a	63	18	1,935.6	MobFox
9e5b41	63	18	1,925.7	MobFox
f56bda	63	17	2,049.3	MobFox
5a6fc0	54	6	5,795.6	MobFox
63fd85	54	6	5,798.7	MobFox
76a7dc	54	6	5,797.3	MobFox
86cd17	54	5	6,956.4	MobFox
94bfa8	4,366	5	8,353.1	Migital
7bb12f	4,392	4	6,717.8	Migital
a3d816	4,381	3	10,000	Migital
807a0a	98	2	0	AppsGeyser
d9162a	4,385	2	16,919.2	Migital
57b67c	56	1	N/A	AppsGeyser
b611ea	4,374	1	N/A	Migital
c7681c	4,416	1	N/A	Migital
d55ece	4,384	1	N/A	Migital
Total	31,257	96		

App cloning

- Market-scale incremental clone detection

Ad fraud

- Include proofs of user viewing or clicking provided by trusted software.

Conclusion



中国互联网安全大会



360互联网安全中心

- Ads are crucial for sustaining mobile app ecosystem.
- Ads involves multiple parties with complex relationship.
- Many security problems affect different parties.
- Measurement is the first step toward solutions.